

Brain × AI 101 Presentation Script

From Biological Neurons to Artificial Networks and Learning

Opening: Landing Page

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Action: Open the landing page and point to the title “Brain × AI 101.” Point to the tagline “A hands-on journey from biological neurons to modern AI.” Then point to the module path labels “Neurons,” “Perception,” and “Learning,” the statistic “86B,” and the time estimates “12 min,” “22 min,” and “26 min.”

Say: Welcome to Brain times AI 101. This course is about one big idea: artificial intelligence may feel new, but many of its ideas are easier to understand when we start with the brain. We will begin with a biological neuron, then move to artificial networks that recognize patterns, and then finish with how learning happens through feedback. The three paths are Neurons, Perception, and Learning. The 86B number reminds us that the human brain has about 86 billion neurons, and each module gives us a short step toward understanding how brains and AI systems process information.

Module 1: Meet the Neuron

A. Introduction

Action: Open Module 1 and show the section labeled “Introduction.” Point to “Selective attention,” then click the flip card scene between the noisy room and bright light scenes.

Say: This first screen starts with something you have probably experienced. You can be in a noisy room where many sounds are happening at once, but if someone says your name, your attention can jump right to it. Your brain is not treating every signal equally. It is filtering, comparing, and giving more importance to signals that matter. When we flip the card to the bright light scene, the idea is similar. A sudden bright signal can stand out because the nervous system is built to notice important changes.

B. Neuron Anatomy

Action: Open “Neuron Anatomy.” Click the glowing hotspots in order: “Dendrites,” “Soma,” “Axon,” and “Terminals.” Use “Next part” between each hotspot, then click “Continue to the sound experiment.”

Say: Now we zoom in on one neuron. Dendrites are the receiving branches. They collect incoming signals from other neurons or from sensors in the body. The soma is the cell body, and it acts like the decision center because it adds signals together. The axon is the long output path. If the signal is strong enough, the message travels down the axon. The terminals are the ending points where the neuron can pass the signal to the next cell.

C. Sound Experiment

Action: Open “Sound Experiment.” Click the input box labeled “Type what Alex hears,” type a sound, click “Send sound,” then try the chips “Alex,” “Everyone listen,” and “Page turning.” Point to “Soma input,” “Threshold 70,” “Stimulate Neuron,” and the PhET simulator controls.

Say: This activity shows how a signal can build up inside a neuron. When we send a sound, the input moves toward the soma. The soma input meter rises as signals arrive. If the total reaches the threshold, the neuron fires. This is called all-or-none firing. That means the neuron either sends a spike or it does not. A stronger sound does not make a half-spike; it helps the total reach the line where the spike happens. The PhET simulator gives us another view of how a neuron responds when it is stimulated.

D. Bridge to AI

Action: Open “Bridge to AI.” Click the four mapping buttons in order: “Receives signals,” “Adds them up,” “Sends one output,” and “Passes it on.” Then click “Continue to Module 2.”

Say: This bridge shows how a biological neuron inspires a simpler artificial neuron. We are not saying an artificial neuron is a perfect copy of a living cell. Instead, we keep the useful pattern. First, signals come in. Next, the system adds them up. Then it sends one output if the evidence is strong enough. Finally, that output can affect the next unit. In AI, those connections have weights, which means some inputs matter more than others.

Module 2: Seeing and Thinking

A. Neural Networks

Action: Open “Neural Networks.” Point to “One Neuron,” then click “Add Hidden Layer ->,” “Add Output Layer ->,” “Go Deeper ->,” and “Reset.”

Say: One artificial neuron can make one weighted decision, but many problems need more than one decision. When we add a hidden layer, several neurons can look at the same input in different ways. When we add an output layer, the network can turn those hidden clues into a final answer. When we go deeper, the network has more layers, so early layers can find simple clues and later layers can combine them into more useful patterns.

B. Activation Functions

Action: Open “Activation Functions.” Move the “Input x =” slider and point to “ReLU output” and “Sigmoid output.”

Say: An activation function is a rule that decides how a neuron responds after adding its input. ReLU turns negative values into zero and lets positive values pass through. Sigmoid squeezes values into a range between zero and one, which can feel like a probability. The important idea is that activation functions bend the behavior of the network. Without that bend, stacking layers would still act like one straight-line rule, and the network would not learn complex patterns well.

C. Neural Selectivity

Action: Open “Neural Selectivity.” Point to “Hubel & Wiesel’s Cat Experiment.” Click cells in the “Input Pattern” grid, click “Show why,” and point to “N1,” “N2,” and “N3.”

Say: This section shows that different neurons can prefer

different patterns. Hubel and Wiesel found that some neurons in the visual cortex responded strongly to lines at certain angles. In this activity, the same input grid is shown to three artificial neurons. They respond differently because their weights are different. One neuron may prefer a vertical pattern, another may prefer a horizontal pattern, and another may prefer a diagonal pattern. When we click Show why, we can see how the input matches each neuron's preferred pattern.

D. CNNs

Action: Open "CNNs." Point to "Seeing in Patches," "Original 3x3," "Padding," "Padded 5x5," "scan with filter," "stride 1," "Output 3x3," and "Filter presets." Click image cells, move the output square, and try the filter presets.

Say: A convolutional neural network, or CNN, looks at an image in small patches. Here the original image is 3×3 . Padding adds a border, making the view 5×5 , so the filter can check edge pixels too. The filter is also called a kernel. It scans one step at a time, and that step size is called stride. With stride 1, the filter moves carefully across the image and creates an output grid. Each output square is a score for one patch.

E. CNN Explainer

Action: Open "CNN Explainer." Show the embedded poloclub iframe and point to "Open CNN Explainer ->" if the embed does not load. Point to the layer-by-layer view and pooling areas.

Say: This embedded explainer lets us watch a CNN process an image layer by layer. Start at the input image, then follow one filter. Early layers often react to simple patterns like edges. Later layers combine those simple parts into bigger visual clues. Pooling reduces the amount of detail, but it keeps the strongest signals. That tradeoff helps the network focus on important features without carrying every pixel forward.

Module 3: Learning to Learn

A. Learning Means Changing

Action: Open "Learning Means Changing." Click "Run Prediction," "Reveal Target," "Measure Error," "Update Weights," and "Test Again."

Say: Learning starts when the model makes a prediction and then checks how wrong it was. First it guesses. Then we reveal the target, which is the correct answer. The difference between the guess and the target is the error. The model uses that error to adjust its weights. After the weights change, the next prediction can become better. Learning means the system changes because of feedback.

B. Three Learning Types

Action: Open "Three Learning Types." Click the cards "Supervised Learning," "Unsupervised Learning," and "Reinforcement Learning."

Say: These three learning types differ by the kind of feedback the model receives. In supervised learning, the model gets the correct answer, like a teacher marking a response. In unsupervised learning, the model does not get labels. It looks for structure, such as groups of similar examples. In

reinforcement learning, the model learns from rewards and penalties. It tries actions and slowly learns which choices lead to better outcomes.

C. Error Goes Down Over Time

Action: Open "Error Goes Down Over Time." Point to "Error by round," then click "Train one round ->" until the model converges. Point to the note about an "epoch."

Say: This chart shows learning across training rounds. Each time we click Train one round, the model makes another pass, checks error, and updates weights. The goal is for the error to go down over time. One pass through the training data is called an epoch. Real models often need many epochs, because improving a model is usually gradual instead of instant.

D. Backpropagation

Action: Open "Backpropagation." Click the four steps "Predict," "Measure Error," "Trace Back," and "Adjust." Use the main button "Start - Make a Prediction" to begin, then continue through the stepper.

Say: Backpropagation answers the question: which weight gets the blame? The model first makes a prediction. Then it measures the error by comparing the prediction with the target. Next, it traces the error backward through the network to find which connections contributed to the mistake. Finally, it adjusts those weights. Think about solving a multi-step math problem and getting the final answer wrong. You would look back through your steps to find where the mistake happened. Backpropagation does something similar for a neural network.

E. Brain $\times$ AI Feedback

Action: Open "Brain $\times$ AI Feedback." Point to "AI model (backprop)," "Brain-inspired model," and "NeuroCorrelation 3D simulation." Click "Start Local App" if available, or click "Open NeuroCorrelation simulation."

Say: Backpropagation is powerful, but the brain does not learn by sending one global error number backward through a neat artificial network. The brain often uses local timing between nearby neurons. A simple way to say this is: neurons that fire together can strengthen their connection. This is called a Hebbian learning idea. The NeuroCorrelation simulation gives us a more brain-like contrast, where connections change based on activity patterns over time.

F. Advanced Learning Labs

Action: Open "Advanced Learning Labs." In "Unsupervised Extension," click "Load Sample" and "Start." In "Supervised Extension," show the TensorFlow Playground iframe. In "Reinforcement Extension," click "Start," "Step Episode," "Q-Learning," and "SARSA."

Say: These labs show the three learning types in a more hands-on way. In the k-means clustering lab, there are no labels, so the system groups points by similarity. In the TensorFlow Playground, supervised learning uses labeled examples to shape a decision boundary. In the reinforcement grid, the agent learns from rewards and penalties. Q-Learning and SARSA are two strategies for updating what the agent believes about actions. The important point is that learning always depends on some kind of feedback signal.

Closing

Course Summary

Action: Open the completion screen. Point to “Biological Neuron,” “Pattern Recognition,” and “Learning to Learn.” Then point to the resources “3Blue1Brown Neural Networks,” “Neural Networks and Deep Learning,” “TensorFlow Playground,” and “Distill.pub.” Finally show “Certificate of Completion,” “Preview Certificate,” and “Download.”

Say: This final screen reviews the course path. In Biological Neuron, we learned how dendrites collect signals, the soma sums them, and the axon fires when the threshold is crossed. In Pattern Recognition, we learned how weights help neurons prefer features like edges and shapes. In Learning to Learn, we learned how feedback can change weights and improve predictions. These four resources are good next steps if you want to go deeper. The certificate section unlocks after the required course steps are complete, then you can preview and download a certificate.

Feedback Request

Action: Ask students to share feedback on science accuracy, example clarity, website flow, and teaching clarity.

Say: Before we finish, I would like your feedback. Please tell me if any science explanation felt inaccurate or confusing. Tell me which examples made the ideas clearer and which examples need work. Also tell me how the website flow felt as you moved from screen to screen. Finally, tell me whether the teaching language felt clear for students who are new to neuroscience and AI. Your feedback helps make the course more accurate, easier to follow, and better for the next group of learners.